

Kaustubh Jeet Mishra

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Professional Summary

Third-year B.Tech student specializing in Artificial Intelligence with hands-on experience building Generative AI and Large Language Model (LLM) applications. Skilled in agentic AI systems, Retrieval-Augmented Generation (RAG), tool calling, conversational memory, and multi-agent orchestration using LangGraph and LangChain. Comfortable working across the LLM application stack – from prompt engineering and embeddings to retrieval pipelines and human-in-the-loop workflows – and interested in building reliable, production-oriented AI systems.

Education

JSS Academy of Technical Education, Noida

2024 – 2028

Bachelor of Technology (B.Tech) in Artificial Intelligence and Machine Learning

Technical Skills

Generative AI / LLMs: LLMs, NLP, Transformers, Self-Attention, Prompt Engineering, Embeddings, Fine-Tuning (LoRA/PEFT), Retrieval-Augmented Generation (RAG)

Agentic AI: AI Agents, Multi-Agent Systems, Tool Calling, Function Calling, ReAct, Agent Architecture, State Management, Memory, Planning, Human-in-the-Loop, Model Context Protocol (MCP)

Frameworks & Libraries: LangChain, LangGraph, Hugging Face, PyTorch, TensorFlow, Keras

Retrieval / Vector Search: FAISS, Semantic Chunking, Document Retrieval Pipelines

LLM APIs / Models: Google Gemini (gemini-2.5-flash), OpenAI (gpt-4o-mini), Groq (GPT-OSS-120B)

ML / Data: Python, Scikit-learn, Tensorflow-keras,

Projects

Teacher–Student AI Classroom [In Progress]

LangGraph, Groq, Gemini, RAG, FAISS, DeepEval

- Building a multi-agent LangGraph voice tutor (Planner, Explainer, Quiz) where students interrupt to ask doubts.
- Built a multimodal RAG pipeline (OCR, Gemini vision, FAISS) with document and web-search tool calling.
- Evaluated retrieval with custom DeepEval metrics; best-first ranking raised ranking precision from 0.68 to 0.90.

Blog Writing Agent

LangGraph, LLMs, Web Search, Streamlit

- Built an autonomous, agentic blog-generation workflow with LangGraph, orchestrating topic ideation, web research, outlining, content generation, human review, and publishing as a multi-step LLM pipeline.
- Integrated real-time web-search tool calling to ground generated content in current information, reducing reliance on model-only knowledge.
- Added human-in-the-loop review checkpoints for iterative feedback before automated formatting and CMS publishing via WordPress/Ghost APIs.

LinkedIn Posting Agent

LangGraph, ChatGroq, LinkedIn API

- Designed an agentic workflow that converts unstructured notes into structured LinkedIn posts using gpt-oss-120b via ChatGroq.
- Implemented human-in-the-loop review allowing generated drafts to be revised or approved before publication.
- Integrated OAuth2 authentication and the LinkedIn API for direct, automated content publishing from the agent.

LangGraph Conversational AI Assistant

LangGraph, LangChain, Gemini, RAG, FAISS

- Developed a stateful conversational AI assistant with persistent memory across chat threads, web search, arithmetic tools, live stock-price retrieval, and external tool execution via function calling.
- Implemented graph-based agent orchestration and state management with LangGraph, powered by gemini-2.5-flash.
- Built a full RAG pipeline (PyPDFLoader, RecursiveCharacterTextSplitter, OpenAI Embeddings, FAISS) for semantic document retrieval and context-grounded question answering.

Professional Strengths

Analytical Thinking | Quality-Driven | Detail-Oriented | Adaptability